Complete Job Lifecycle — From Prospect to Installed Customer

Audience: All roles — the operational overview document for understanding how a Lumina job flows from first prospect visit through installed, live customer.

This is the master reference for the entire job pipeline. Use it to understand who owns each stage, what messages fire automatically, and how to recover from common failure modes. Permanent residential and commercial lighting that works as hard as you do — delivered by a pipeline that communicates with the customer at every milestone so no one ever has to chase for status.

What you'll need

- Access to your role's Lumina interface (Sales Mode, Installer Mode, Dealer Dashboard, or Corporate Dashboard)
- Familiarity with your role's specific guide (Sales Mode Guide, Day 1 Electrician Guide, Day 2 Install Guide, Dealer Dashboard Guide, Corporate Dashboard Guide)
- If you're debugging a stalled or broken job: dealer admin access or backend access

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1. The full pipeline at a glance

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│                                                                  │
│   PROSPECT                  INSTALL                  CUSTOMER    │
│   (Salesperson)             (Field crews)            (Live)      │
│                                                                  │
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  │ Draft  │──▶│ Estimate │──▶│ Signed │──▶│ Pre-wire │──▶│ Pre-wire │
  │        │    │   Sent   │    │        │    │Scheduled │    │ Complete │
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```

Status	Owner	Plain English
Draft	Salesperson	Estimate is being built but not generated.

Estimate Sent	Salesperson	Estimate generated but customer hasn't signed. *(Used in legacy flows; the wizard usually goes straight to Signed.)*
Signed	Day 1 electrician	Customer signed. Job is officially booked.
Pre-wire Scheduled	Day 1 electrician	Day 1 has a date.
Pre-wire Complete	Day 2 install team	Wires are run. Ready for Day 2.
Install Scheduled	Day 2 install team	Day 2 has a date.
Install Complete	Customer	Job is closed. Customer has the lights.

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2. Stage-by-stage walkthrough

For each stage:

- **Owner** — whose responsibility it is to advance the job
- **What the user does** — the action that moves the job forward
- **What gets created/updated** — data side effects
- **Automated messages fired** — what the customer receives
- **Customer experience** — what they see from their side

Stage 1 — Prospect → Draft

Owner: Salesperson

What the user does:

1. Salesperson opens Sales Mode and taps **New Visit**.
2. Fills in the **Customer Info** form (name, email, phone, address, optional referral code, notes, photos).
3. Taps **Continue to zones** →.

What gets created:

- A new sales job in the dealer's collection with auto-generated job number (`NXG-{date}-{seq}`)
- The job has status **Draft**, embedded prospect record, dealer code, salesperson identifier, creation timestamp
- Any home photos uploaded are stored against the job
- If a valid referral code was entered, the referral pipeline is updated to "visit scheduled"

Automated messages: None. The customer is unaware so far.

Customer experience: Sitting with the salesperson. Nothing automated has fired yet.

Stage 2 — Draft → (still Draft, building estimate)

Owner: Salesperson

What the user does:

The salesperson walks through the 5-step Estimate Wizard:

1. **Home Photo** — captures the blueprint background photo
2. **Controller Placement** — describes mount location, interior/exterior, distance to outlet, optional photo
3. **Channel Setup** — adds one or more LED channel runs
4. **Power Injection Points** — adds injection points for runs over 100 ft
5. **Summary and Review** — reviews everything and taps **Generate Estimate**

What gets updated:

- The job is updated continuously as the wizard progresses (home photo URL, controller mount, channel runs, injection points)
- On **Generate Estimate**, the system loads dealer pricing and writes a full **EstimateBreakdown** (line items by category, subtotals, margin)
- Job is still **Draft**

Automated messages: None.

Customer experience: Sitting with the salesperson. Watches the estimate build live on the device.

Stage 3 — Draft → Signed

Owner: Salesperson (handing the device to the customer)

What the user does:

1. Salesperson taps **Sign** from the estimate preview
2. Hands the device to the customer
3. Customer reviews the estimate, draws a signature, and taps **Approve & confirm**

What gets updated:

- Customer signature PNG is uploaded to the job's cloud folder
- Job status → **Signed (** `estimateSigned` **)**
- `estimateSignedAt` timestamp set
- `customerSignatureUrl` populated
- If a referral code was used, the referral pipeline moves to "confirmed" and the referrer's credit is locked in
- The job becomes visible in the dealer's Day 1 Queue

Automated messages fired:

- **Booking Confirmation Email** to the customer (subject: "*You're booked with Nex-Gen LED!*") — explains the 2-day process, access requirements, and dealer contact. *(Toggleable per dealer.)*

Customer experience: Signs and immediately sees confirmation. Within seconds, the booking email arrives.

Stage 4 — Signed → Pre-wire Scheduled

Owner: Day 1 electrician

What the user does:

1. Electrician opens the Day 1 Queue
2. Finds the new signed job
3. Taps **Schedule Day 1**
4. Picks a date

What gets updated:

- Job status → **Pre-wire Scheduled (** `prewireScheduled` **)**
- `day1Date` set

Automated messages fired:

- **Day 1 Confirmation SMS** to the customer:

  > *"Hi {first name}, your Nex-Gen LED prep day is confirmed for {date}. Our technician will handle the wiring — no lights go up this day, that's Day 2. Questions? Reply here. — {dealer sign-off}"*

Customer experience: SMS confirming the Day 1 date.

Stage 5 — Pre-wire Scheduled → (Day 1 visit happens)

Owner: Day 1 electrician

What the user does:

1. **The night before** the visit (6:00 PM Central), the system automatically sends the **Day 1 Reminder SMS**
2. On the day of the visit, the electrician arrives, opens the **Day 1 Blueprint**, and works through every task
3. Each task is checked off in the app as completed

Automated messages fired during this stage:

- **Day 1 Reminder SMS** (the night before, automatic) — reminds the customer about access requirements. *(Toggleable per dealer.)*

What gets updated:

- `day1CompletedTaskIds` array updated as tasks are checked
- No status change yet — that happens at check-out

Customer experience: Gets the reminder SMS the night before. On the day, the electrician shows up and runs wiring. No lights yet.

Stage 6 — Pre-wire Scheduled → Pre-wire Complete

Owner: Day 1 electrician

What the user does:

1. Completes all tasks on the Day 1 Blueprint, including the verification task ("Verify all wires are accessible outside wall for Day 2")
2. Taps **Mark Day 1 complete**
3. Confirms by entering their name in the dialog

What gets updated:

- Job status → **Pre-wire Complete (** `prewireComplete` **)**
- `day1CompletedAt` timestamp set
- `day1TechUid` recorded
- Job disappears from Day 1 Queue, appears in Day 2 Queue

Automated messages fired:

- **Day 1 Complete SMS** to the customer:

> *"Great news, {first name}! Wiring prep is complete at your home. Your light installation day is coming soon — we'll text you the night before. — {dealer sign-off}"*

Customer experience: "Wiring done" SMS the same day.

Stage 7 — Pre-wire Complete → Install Scheduled

Owner: Day 2 install team

What the user does:

1. Day 2 installer opens the Day 2 Queue
2. Finds the new pre-wire complete job
3. Taps **Schedule Day 2**
4. Picks a date (must be on or after the Day 1 completion date)

What gets updated:

- Job status → **Install Scheduled (** `installScheduled` **)**
- `day2Date` set

Automated messages fired:

- **Day 2 Confirmation SMS** to the customer:

  > *"Hi {first name}, your Nex-Gen LED light installation is confirmed for {date}. Our install team will have your lights up and running. Get excited! — {dealer sign-off}"*

Customer experience: SMS confirming the install date.

Stage 8 — Install Scheduled → (Day 2 visit happens)

Owner: Day 2 install team

What the user does:

1. **The night before** the visit (6:00 PM Central), the system automatically sends the **Day 2 Reminder SMS**
2. On the day, install team arrives, opens the Day 2 Blueprint, works through every install task

Automated messages fired:

- **Day 2 Reminder SMS** (the night before, automatic). *(Toggleable per dealer.)*

Customer experience: Gets the reminder SMS, then watches the install team put up the lights.

Stage 9 — Day 2 Wrap-Up Sequence (the most important step)

Owner: Day 2 install team

What the user does:

After all install tasks are checked, the installer taps **Wrap up install** → and walks through the 4-step wrap-up:

Step 1 — Install Photos

- Captures one completion photo per channel
- Photos uploaded to the job's cloud folder
- Step is gated — cannot proceed without photos for every channel

Step 2 — Material Check-In

- Records returned quantity for each material in the estimate
- App auto-calculates used quantity and waste percentage
- Data persisted to the job's `actualMaterialUsage` field
- Feeds the dealer's Waste Intelligence dashboard

Step 3 — Customer Account Creation

- Installer confirms the customer's name and email
- Taps **Create account**
- The system:
  - Creates a new Firebase Auth user
  - Generates a password reset link
  - Sends the **Account Setup Email** (subject: "*Set up your Nex-Gen LED Lumina account*") with the reset link and app download buttons
  - Creates a `/users/{uid}` document tagged as the customer's primary account
- The new UID is saved on the job as `linkedUserId`

Step 4 — Close Job

- Installer reviews the summary card (photos captured, materials checked in, account created)
- Taps **Finish & Close Job** and confirms in the dialog

What gets updated at close:

- Job status → **Install Complete (`installComplete`)**
- `day2CompletedAt` timestamp set
- `day2TechUid` recorded
- Job disappears from the Day 2 Queue

Automated messages fired at close:

- **Install Complete Email** to the customer (subject: "*Your Nex-Gen LED system is live!*") — confirms the install, provides app download links, references the account setup email, and signs off with the dealer name. *(Toggleable per dealer.)*

Customer experience:

1. While the installer is on Step 3, the customer gets the **Account Setup Email** with their password link.
2. After Step 4 closes, the customer gets the **Install Complete Email** with download links.
3. Customer downloads Lumina, sets their password, signs in, controls their lights for the first time.

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3. Stage summary — who owns what and what fires

#	Status	Owner	Trigger	Customer messages
1	**Draft**	Salesperson	Wizard in progress	None
2	**Estimate Sent**	Salesperson	(legacy)	None
3	**Signed**	Day 1 electrician	Customer taps **Approve & confirm**	Booking Confirmation Email
4	**Pre-wire Scheduled**	Day 1 electrician	Electrician picks Day 1 date	Day 1 Confirmation SMS

5	*(during wait)*	Day 1 electrician	Cron at 6pm Central night before	Day 1 Reminder SMS
6	**Pre-wire Complete**	Day 2 install team	Electrician taps **Mark Day 1 complete**	Day 1 Complete SMS
7	**Install Scheduled**	Day 2 install team	Installer picks Day 2 date	Day 2 Confirmation SMS
8	*(during wait)*	Day 2 install team	Cron at 6pm Central night before	Day 2 Reminder SMS
9	*(during wrap-up)*	Day 2 install team	Installer taps **Create account** in Step 3	Account Setup Email
10	**Install Complete**	Customer	Installer taps **Finish & Close Job**	Install Complete Email

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4. Technical integration notes

For technical operators. Field crews can skip this.

Firestore documents touched at each stage

Stage	Documents
Draft creation	`sales_jobs/{jobId}` created with embedded prospect; optional referral pipeline updated
Wizard progress	`sales_jobs/{jobId}` updated continuously (home photo URL, controller mount, channels, injections)
Estimate generation	`sales_jobs/{jobId}.estimateBreakdown` written
Signature	Signature PNG uploaded to Cloud Storage; job status → `estimateSigned`; signature URL written
Day 1 schedule	`sales_jobs/{jobId}.day1Date` + status → `prewireScheduled`
Day 1 complete	`sales_jobs/{jobId}.day1CompletedAt`, `day1TechUid`, status → `prewireComplete`
Day 2 schedule	`sales_jobs/{jobId}.day2Date` + status → `installScheduled`
Wrap-up Step 1	`sales_jobs/{jobId}.installCompletePhotoUrls` (positional list, 1 per channel)
Wrap-up Step 2	`sales_jobs/{jobId}.actualMaterialUsage` map (per item: estimated, returned, used, waste %)
Wrap-up Step 3	New Firebase Auth user; new `users/{uid}` document; `sales_jobs/{jobId}.linkedUserId`
Wrap-up Step 4	`sales_jobs/{jobId}.day2CompletedAt`, `day2TechUid`, status → `installComplete`

Cloud functions

Function	Trigger	Purpose
onSalesJobStatusChanged	Firestore update on `sales_jobs/{jobId}`	Detects status transitions and Day 1 completion; sends Booking Confirmation Email, Day 1/Day 2 confirmation SMS, Day 1 complete SMS, and Install Complete Email
sendInstallReminders	Scheduled cron at 6pm Central daily	Queries jobs whose Day 1 or Day 2 date is tomorrow (Central time), sends reminder SMS to each
createCustomerAccount	Callable from Day 2 wrap-up Step 3	Creates Firebase Auth user, sends Account Setup Email, seeds user profile document

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5. Common failure points and how to recover

Things go wrong. Here's how to fix them without losing customer data.

Failure 1 — Customer phone number is invalid

Symptom: SMS reminders bounce or never arrive.

What to do:

1. Open the job in the dealer's My Estimates list
2. Find the customer's phone field on the prospect info section
3. Correct the number using the dealer admin's edit-job flow
4. Subsequent SMS uses the corrected number. Previous failed messages will not auto-retry.
5. If the customer missed a reminder because of this, contact them directly to confirm the appointment.

Failure 2 — Day 1 check-out is blocked because tasks are incomplete

Symptom: Mark Day 1 complete is grayed out and reads "*Complete all tasks to mark complete*".

What to do:

1. Scroll up through the task checklist on the Day 1 Blueprint
2. Look for any task without a green checkmark
3. Common culprit: the verification task ("Verify all wires are accessible outside wall for Day 2") at the bottom — easy to miss
4. Check the missing task(s)
5. The button activates

If the electrician genuinely can't complete a task (a pre-run wire is missing, for example), do **not** check it off falsely. Instead:

- Document the issue with the dealer admin
- Leave the job in `prewireScheduled`
- Schedule a follow-up Day 1 visit

Failure 3 — Customer account creation fails during wrap-up

Symptom: Step 3 of wrap-up shows an error, no account created, no email sent.

Possible causes:

- Email already exists (the customer has a previous Lumina account)
- Email format invalid
- Network failure
- Backend service issue

What to do:

1. **If the email already exists:** Tap **Re-send invite** to send a fresh password reset link to the existing account. If the customer doesn't recognize the existing account, contact the dealer admin to investigate.
2. **If the format is invalid:** Verify the email with the customer, edit it on the prospect record, try again.
3. **If you can't resolve on-site:** Continue through Step 4 to close the job. The customer won't receive their account setup email automatically — contact your dealer admin and have them manually provision the account afterwards.

Failure 4 — Job is stuck in a status

Symptom: A job sits in one status for many days. Customer or dealer wants to advance it.

Common scenarios:

- A job is **Signed** but no electrician has picked it up
- A job is **Pre-wire Complete** but no installer has scheduled Day 2
- A job somehow got into the wrong status (manual data correction needed)

For genuinely stalled jobs:

1. Open the job in the dealer's **My Estimates** list (or for corporate, the Pipeline tab)
2. Tap the job to open the **Job Detail** screen
3. The Job Detail screen has **manual status advance buttons** that appear depending on the current status:
   - **Signed** → "Schedule pre-wire" (advances to `prewireScheduled`)
   - **Pre-wire Scheduled** → "Mark pre-wire complete" (advances to `prewireComplete`)
   - **Pre-wire Complete** → "Mark install complete" (advances to `installComplete`)
   - **Install Scheduled** → "Mark install complete" (advances to `installComplete`)
4. Tap the appropriate button to manually move the job forward

> ***Important:*** *Manual advances still trigger the corresponding automated messages. Use this only when you actually want the customer to receive the next message — don't manually advance a job whose physical work hasn't actually been done.*

Failure 5 — Wrong photo or info on the prospect

Symptom: Salesperson typed something incorrectly or attached the wrong photo.

What to do:

1. Open the job from My Estimates
2. Resume the wizard or use the dealer admin's edit flow
3. Correct the field, save
4. If the job is still in **Draft**, no automated messages have fired — edit freely
5. If the job is **Signed** or beyond, edits don't re-trigger prior messages — they only affect future messages and the install crews' view of the job

Failure 6 — Installer marks Day 1 or Day 2 complete by accident

Symptom: Job is at `prewireComplete` or `installComplete` but the work isn't actually done.

What to do:

1. Contact the dealer admin immediately
2. The dealer admin can manually correct the job status using backend tools
3. **The customer has already received the next-stage SMS or email.** Reach out directly to clarify
4. Re-do the work with the proper check-out flow

This is one of the most damaging errors because it triggers customer messages that don't match reality. Train installers to double-check before tapping the complete button.

Failure 7 — Reminder SMS didn't arrive the night before

Symptom: Customer says they didn't get the reminder.

Possible causes:

- The dealer disabled the reminder toggle
- The customer's phone is wrong
- The job's date wasn't actually scheduled
- SMS provider hiccup

What to do:

1. Check the dealer's messaging configuration — confirm the relevant reminder toggle (Day 1 or Day 2) is on
2. Verify the customer's phone number on the job
3. Verify the job's `day1Date` or `day2Date` is correct
4. If everything's correct, this was probably a one-off provider hiccup. Send a manual reminder via your usual support channels and confirm the appointment with the customer directly.

Failure 8 — Photo upload fails during the wizard or wrap-up

Symptom: A photo capture fails to upload.

What to do:

1. Check the device's network connection
2. Retry the capture
3. If upload still fails, save the photo to the device gallery and try uploading from there
4. If Wrap-up Step 1 (install photos) is the blocker, you cannot proceed past Step 1 without a photo for every channel — keep trying until they upload

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6. The customer's end-to-end experience

From the customer's point of view:

Day	Event	Channel
Day 0 (sales visit)	Salesperson visits and walks through the wizard live	In person
Day 0	Customer signs the estimate on the device	In person
Day 0 (within seconds)	Booking confirmation email arrives	Email

Day X (some days/weeks later)	Day 1 confirmation SMS arrives	SMS
Day X − 1 (evening)	Day 1 reminder SMS arrives	SMS
Day X (during visit)	Electrician arrives and runs all wiring	In person
Day X (after visit)	Day 1 complete SMS arrives	SMS
Day Y (some days later)	Day 2 confirmation SMS arrives	SMS
Day Y − 1 (evening)	Day 2 reminder SMS arrives	SMS
Day Y (during visit)	Install team arrives and the lights go up	In person
Day Y (during wrap-up)	Account setup email arrives	Email
Day Y (after visit)	Install complete email arrives	Email
Day Y (after install)	Customer downloads Lumina, sets password, controls their lights	App

The whole experience is designed so the customer is always informed and never has to chase the dealer for status updates.

What success looks like

- Jobs move through the pipeline at a steady cadence — no status sitting more than 7–14 days unattended
- Every automated message fires on its intended stage transition, and the customer sees confirmations arrive within seconds
- The night-before reminders reduce missed appointments to near zero
- Wrap-up Step 3 creates the customer account on the first try, and Step 4 closes the job cleanly
- By the time the installer leaves, the customer has downloaded Lumina, set their password, and toggled their lights at least once

If something isn't working

Start with the failure point that matches the symptom — they're all covered in Section 5. The short version:

Customer not getting messages? Check phone/email on the job, then the dealer's messaging toggles. **Job stuck in a status?** Use the manual advance buttons on the Job Detail screen (but only when the physical work has been done). **Account creation failing?** Look for an existing Lumina account; use **Re-send invite** if found. **Wrong data on the prospect?** Edit via the dealer admin flow. Past messages don't retrigger; future ones use the corrected data. **Installer marked complete by accident?** Contact the dealer admin immediately, reach out to the customer, re-do the check-out properly.

If a failure isn't covered in Section 5, contact your Nex-Gen LED LLC corporate contact.

Need help? This document is the master reference for the job pipeline. For role-specific guides, see the Sales Mode Guide, Day 1 Electrician Guide, Day 2 Install Guide, and Corporate Dashboard Guide.